
User Interface Design and Usability Engineering

Y. Raghu Reddy

Software Engineering Research Centre
IIIT Hyderabad, India



Simple Exercise...

Design a User Interface for a product that does Addition, Subtraction, Multiplication, Division of two natural numbers.

List down “assumptions” you made (if any) while designing this user interface !

Who is the user ?



Persona !

- What's a persona ?
- Why is it important to understand Persona ?
- Who are the kinds of people do you expect to use your product?
- Users' needs, expectations, behaviors, and goals

In your group, can you come up with the persona that you are targeting and then re-design the UI

Persona !

- About Me:
 - Interests
 - Goals
 - Other Characteristics (Attention span, Imagination, Digital fluency, Imagination, NUMERACY, etc)
-

Persona Example !



RYLEE CHILD

ABOUT ME

AGE: 5 years / Prep
GENDER: Female
FAMILY: Parents, Brother, Dog
LOCATION: Melbourne, Victoria
LANGUAGE: English

INTERESTS

Music
Books
Colouring-In
Dolls
Painting

CHARACTERISTICS

Attention Span:	
Literacy:	
Imagination:	
Digital Fluency:	
Numeracy:	

GOALS

- To learn how to read
- To get better at Art

Another activity...

As a team, pick one app you love and one you hate !

Write the reasons...

What happens outside (Customer?) and What happened inside (Engineering?)



IS

STARTING AT \$37,325*
2011 40P33L46

BUILD CURRENT OFFERS



RS

STARTING AT \$38,100*
268 HP 35LV6

BUILD CURRENT OFFERS



ES HYBRID

STARTING AT \$41,020*
40 MPG COMBINED PUTING*

BUILD CURRENT OFFERS



GS

STARTING AT \$45,615*
2011 40P35LV6

BUILD CURRENT OFFERS



GSHYBRID

STARTING AT \$36,080
338 TOTAL SYSTEM HP*

BUILD CURRENT OFFERS



GSF

STARTING AT \$38,440*
467 HP 50LV8

BUILD CURRENT OFFERS



LS

STARTING AT \$72,520
366 HP 46LV8 19WD

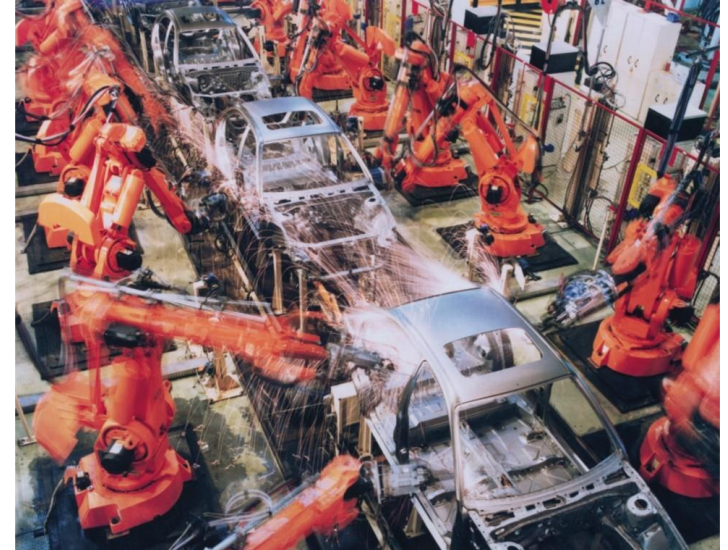
BUILD CURRENT OFFERS



LSHYBRID

STARTING AT \$70,440 43
810 TOTAL SYSTEM HP*

BUILD CURRENT OFFERS



UI

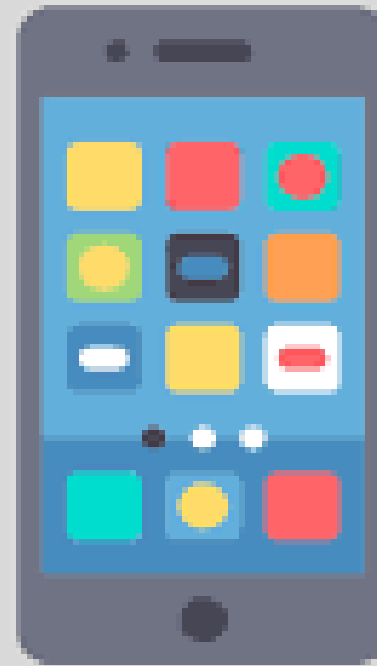
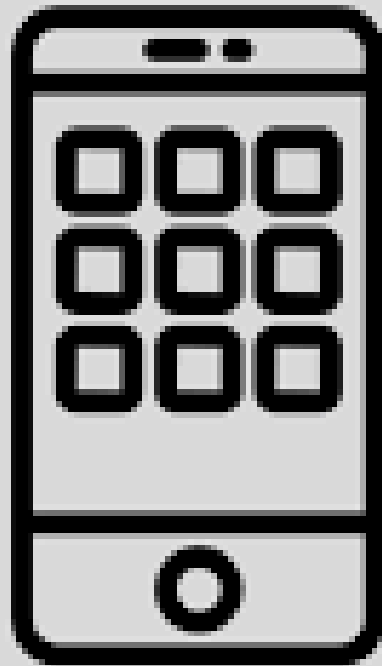


UX



UX

users
usability
wireframe
navigation
analysis



UI

color
content
branding
layout
visual

Let's repeat the activity...

As a team, pick one app you love and one you hate specifically from a UI Design perspective !

Write the reasons...

Let's do an activity !

Are U Ready?

1



2



3



4



5



6



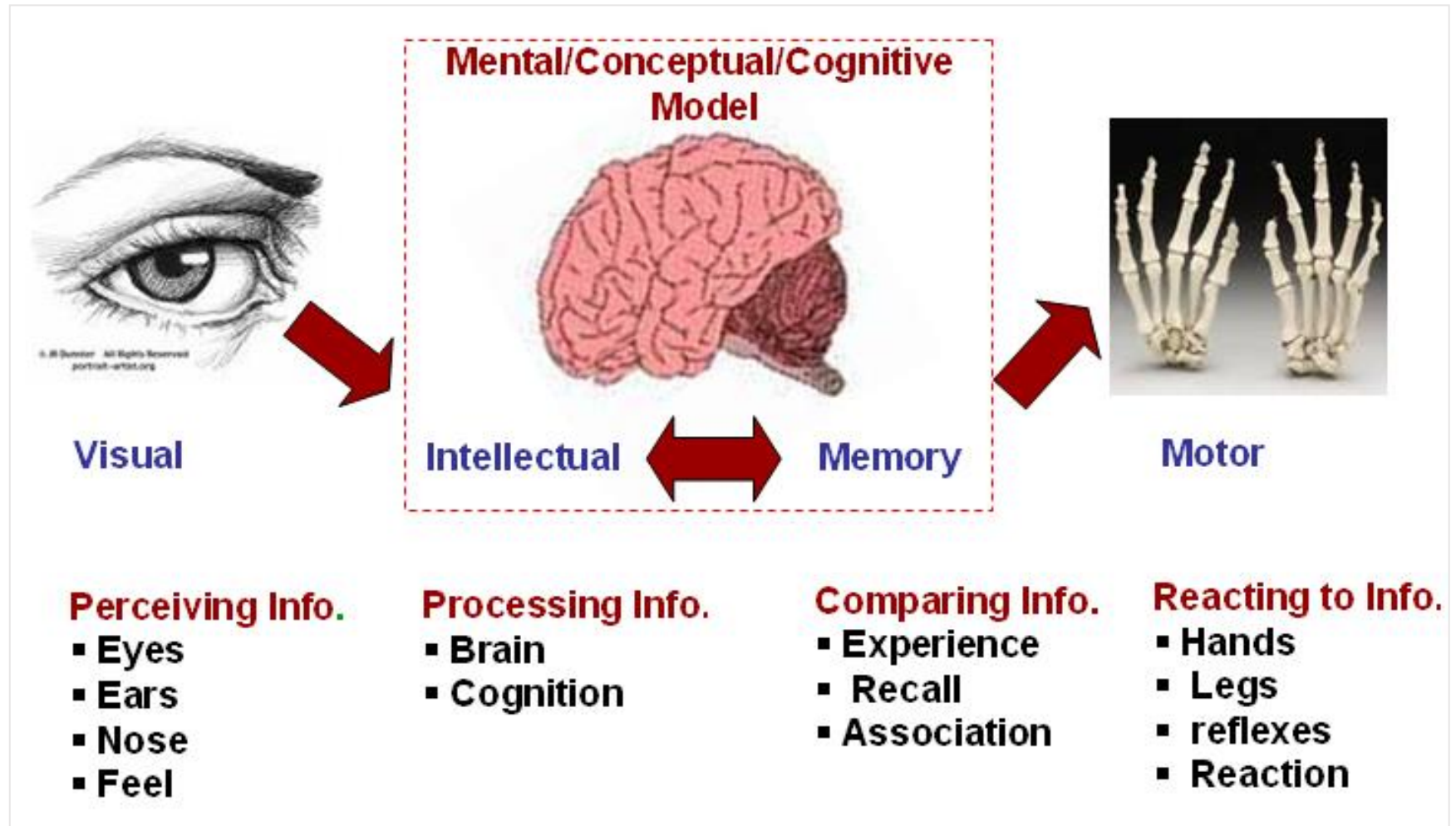
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8



Human Interaction with Outside World !



Task Flow (Cognitive Model)

Example: Accessing & Reading Email on Mobile



Mental Models

- A mental model is what a user *believes* they know about how a system works based on past experiences.
- **Jakob's Law:** Users spend most of their time on other websites, meaning they expect your product to work similarly to what they already know.
- **Belief vs. Fact:** Models are not based on technical facts but on internal simulations used to predict outcomes.

Why It Matters for UI Design

- **Reduces Cognitive Load:** When a UI aligns with existing models, users don't have to "relearn" how to use your tool, saving mental effort.
- **Predicts Behavior:** Mental models allow users to predict the outcome of an interaction before they even click.
- **Eliminates Friction:** Mismatches lead to "user error," frustration, and high abandonment rates.

Universal UI Metaphors & Patterns

- **Skeuomorphism:** Digital elements that mimic their real-world counterparts (e.g., a notebook icon that looks like paper) to bridge the gap.
- **Affordances & Signifiers:** A "cart" icon signals a place to store items for purchase; a "magnifying glass" signifies search functionality.
- **Standard Placement:** Users expect "Settings" at the top-right and "Navigation" at the top or left.

Bridging the Designer-User gap

- **The Mismatch:** Designers often have a complex "expert model" that differs significantly from the novice user's model.
- **Research Tools:** Use **Card Sorting** or **Think-Aloud Testing** to discover how users naturally categorize and interpret information.
- **Strategizing Innovation:** Only break a mental model if the new solution provides significant value; otherwise, stick to familiar patterns to ensure a seamless experience.



11 Reasons Mobile Apps Fail

The mobile industry is rapidly growing to an endless scope. But increased competition also leads to increased chances of failure.



by Lauren Gilmore 8 min · Jul. 05, 18 · Mobile Zone · Opinion

User Issues

There's nothing more frustrating than a mobile app that crashes - or no performance - is the number one reason why users

6. Bad User Experience

Offering a compelling user experience is crucial for the success of any app. If the available features are key, user interface design is a major factor. A poor user interface is difficult to use and causes performance issues.

Solution: Deviate enough attention to UX/UI best practices

testbytes

Making Quality a Habit

Top 8 Reasons for App Failure and How to Avoid Them

Tuesday August 7, 2018

App Testing

5. Poor User Interface

Generally, poor user interface means a poor design. Today, most apps are rejected by users ultimately leading to their failure due to poor user experience. If the performance of an app is unsatisfying to the users, they will uninstall it in no time. If users are unable to perform basic functions seamlessly, then it will be very hard for developers to sell their product. There are various reasons how an app can give poor user interface such as:

Product Failures

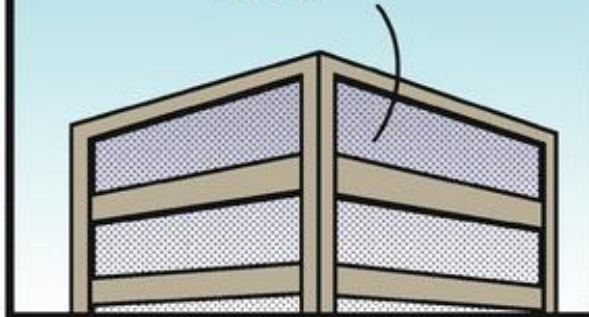


CUSTOMERS ARE
COMPLAINING BECAUSE
OUR USER INTERFACE
IS CONFUSING.



DILBERT.COM @SCOTTADAMSSAYS

FOR EXAMPLE, OUR
MENU CHOICE FOR
DELETING A FILE
IS LABELED "SAVE
FILE."



5-2-18 ©2018 Scott Adams, Inc./Dist. by Andrews McMeel

THAT'S
WHY WE
HAVE A
HELP
MENU.



OUR HELP
MENU IS
LABELED
"REFORMAT
HARD
DRIVE."



Design of User Interfaces

- Inside-out design :

Develop a system, then add the interface

- Outside-in design :

Design the interface, then build the system to support it

When decisions are made, either the **developer must conform to the user**, or the user must conform to the developer.

Traditional CS is entirely inside out

Why is Designing UIs Hard ?

- Good UIs take time to design
 - Many programmers are lazy
- Designing a good UI requires thinking like the user instead of an engineer
 - Engineers often think they are users
- Different users want different things

Engineers love features

They want to do everything the technology allows!



Why is Designing UIs Hard ?

- Designers are usually experts
 - They view all functions as having equal weight
- Marketing want UIs designed for beginners
 - They sell to beginners
 - Many have only novice semantic knowledge
- Most users are in between—intermediates !

Don't weld on training wheels

Understand the Users

- Work experience
- Computer experience
- Age
- Education
- Reading skills
- Language skills
- Work environment
- Task frequency
- ... many more possibilities

It is important to
know **who** the user is

Distribution of Users

Beginners

What does the application do?

What doesn't it do?

How do I print?

Where do I start?

Intermediates

Forgot how

Where is it?

Remind me ...

Can I undo?

More features?

Experts

Automate?

Shortcuts?

Can I change?

Customize?

Keyboard shortcut?

Dangerous?

UI Patterns

<http://ui-patterns.com/patterns>

Usability

ISO (ISO/IEC 9126-1:2001) defines usability as
“The extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use”

UI Vs Usability

UI

Focuses on look & feel

Visual aesthetics

Designer-driven

“Is it attractive?”

Usability

Focuses on task success

Effectiveness &
efficiency

User-driven

“Can users do what they
want?”

Bad Usability Examples



A beautiful UI can still be unusable !

Usability

Attributes for Evaluating Usability

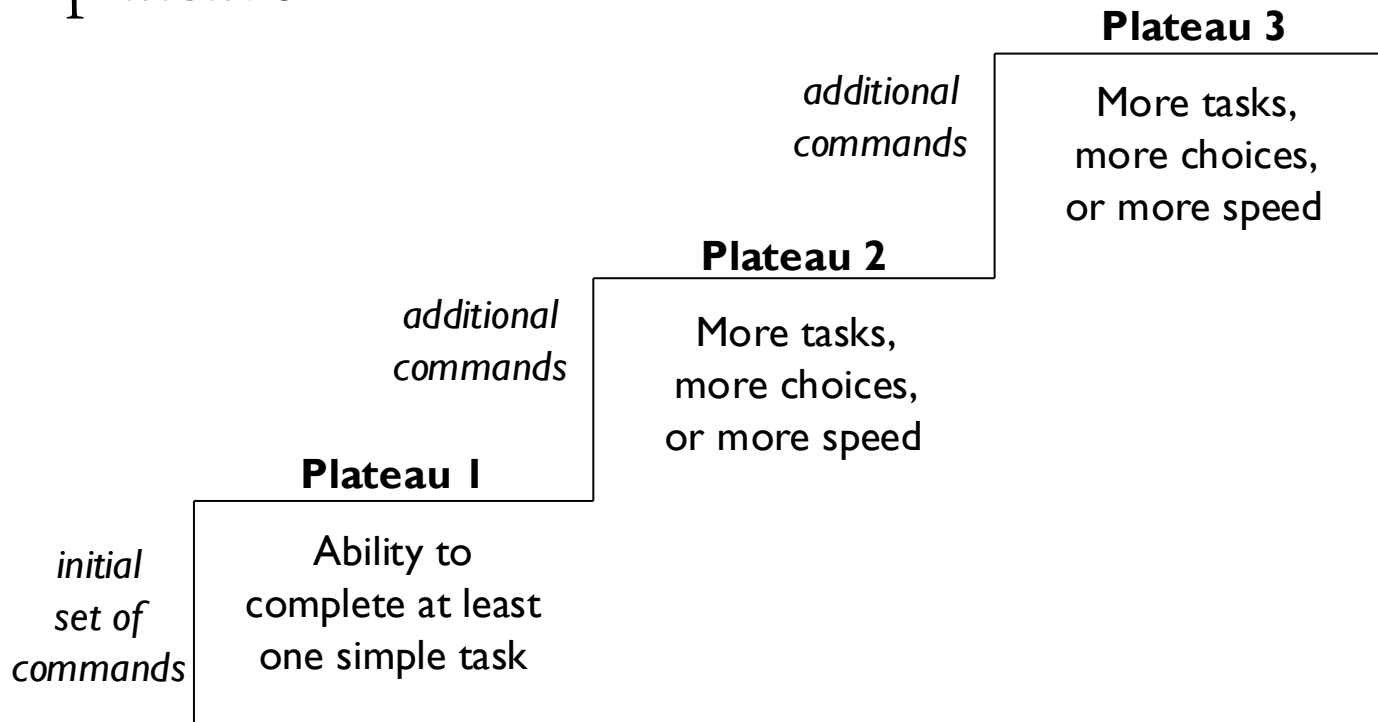
- i. Learnability
- ii. Efficiency
- iii. Memorability
- iv. Errors
- v. Satisfaction



These attributes often conflict (e.g., learnability vs efficiency). Usability design is about balancing trade-offs.

1. Time to Learn (Learnability)

- How long it takes to learn how to use an interface
- With complicated interfaces, learning happens in “plateaus”



2. Speed of Performance (Efficiency)

- Speed of user interface, **NOT** software
 - Number of characters to type, buttons to press, mouse-clicks, mouse movements, ...
 - Speed of performance often directly conflicts with time to learn
 - That is, faster systems are often harder to learn
 - Command lines vs. GUIs
-

3. Rate of User Errors

- A UI needs to be designed in such a way that user mistakes are less likely
 - Affected by factors such as :
 - Consistency
 - Instructions
 - Logical arrangement of screens
 - Importance depends on criticality of software
-

4. Retention of Skills (Memorability)

- We quickly forget how to use some user interfaces, but remember others for life
 - Calculus vs. algebra
 - Airplanes vs. bicycles
 - Affected by how closely the syntax of the operations match our understanding
 - If learning is very fast, retention may be less important
-

5. Subjective Satisfaction

- How comfortable the users are with the software
 - Previous criteria are very analytical, objective, and measurable
 - Subjective satisfaction captures other issues that are more specific to individual taste and background
 - A little harder to measure
-

Let's revisit the paper plane

Learnability (Ease of Assembly)

Was the design intuitive? Did you struggle to roll the paper or attach the fins

Efficiency of Use

A well-designed rocket requires less force (lower input effort) to achieve a greater distance.

Memorability

Can you build another one if you came back 2 years later that gives similar performance?

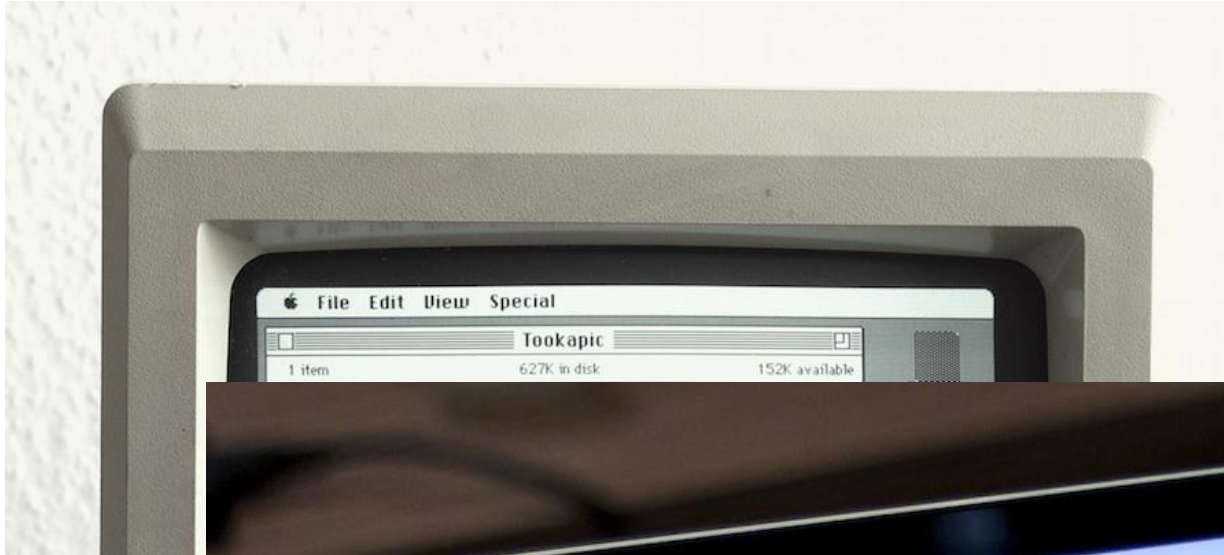
User Satisfaction (Subjective Measure)

Did the design feel secure, was it easiest to build, was it the the most "polished"?

Shneiderman's Golden Usability Principles

1. Strive for consistency
 2. Design usable and discoverable shortcuts
 3. Provide appropriate feedback
 4. Yield closure
 5. Provide appropriate error handling
 6. Allow users to reverse (undo) all actions
 7. Support internal locus of control
 8. Reduce the Short Term Memory (STM) load
-

Principle (1/8) - Consistency



MacOS menu bar
remains consistent!
1980s to 2019 !



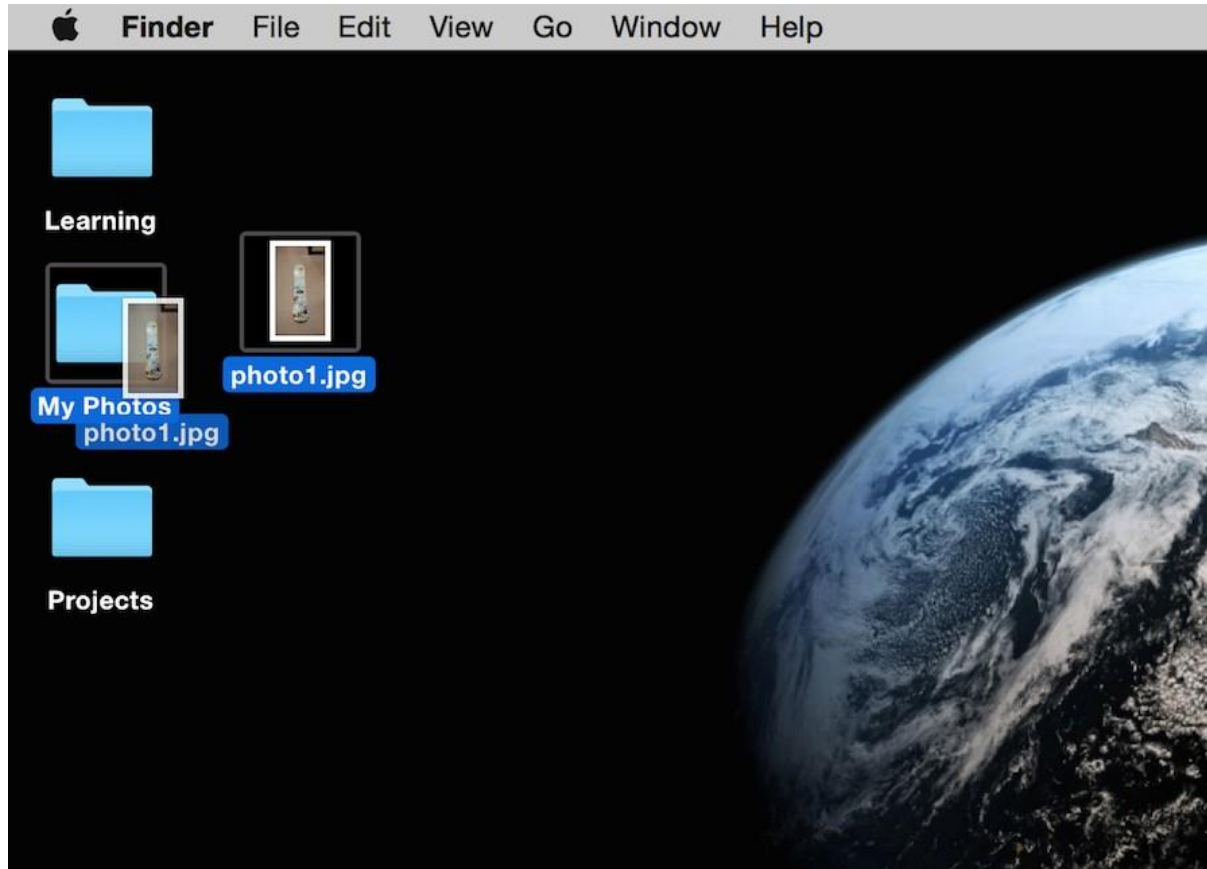
Various aspects of UI Must be consistent ! Changes break Consistency!!

Principle (2/8) – Shortcuts

Design usable and discoverable shortcuts

- ❑ Users must be able to find them
 - ❑ Users must be able to remember them
-
- Copy and Paste (Command-C and Command-V)
 - Screenshots (Command-Shift-3)

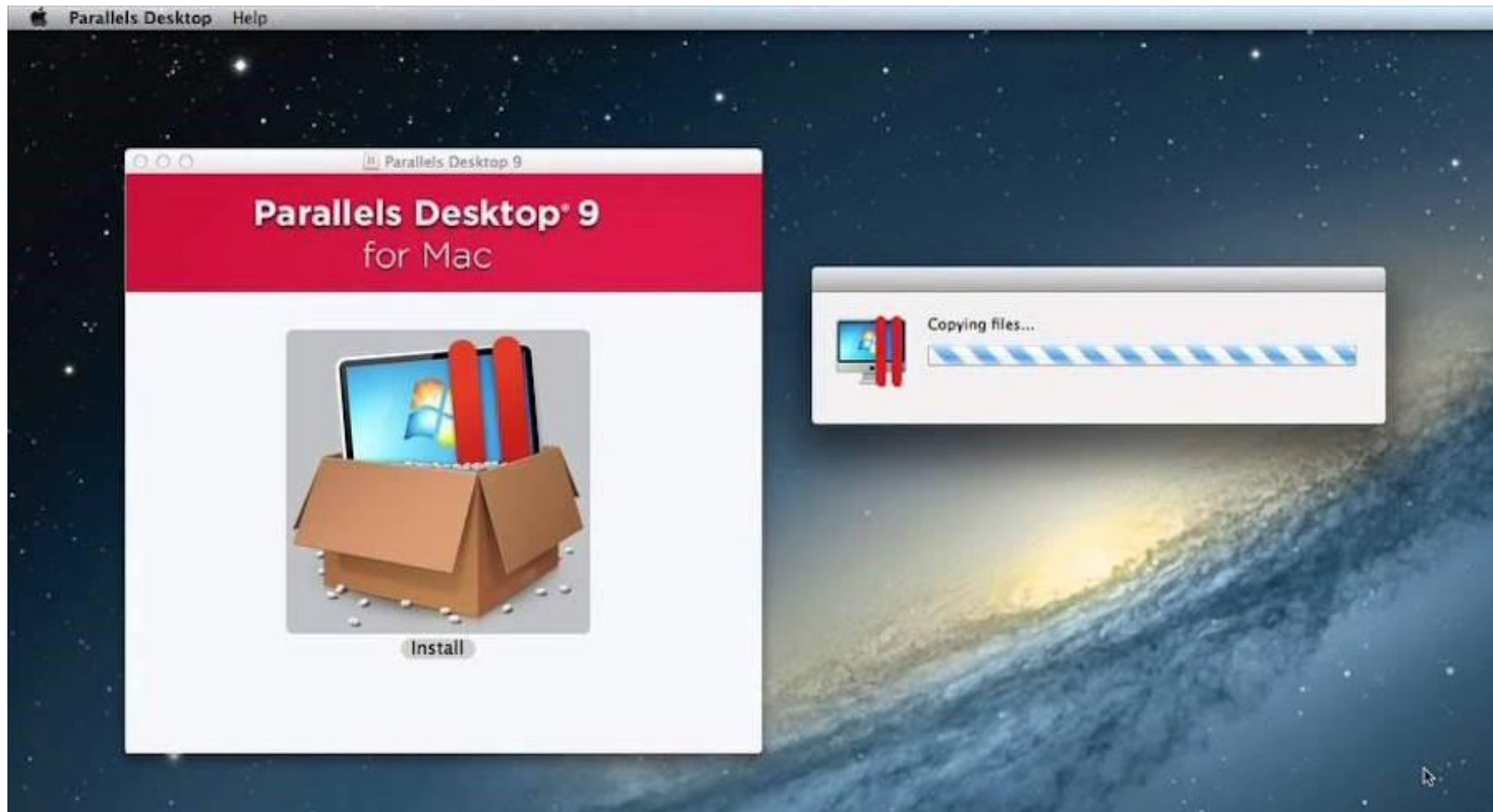
Principle (3/8) - Feedback



Folder is shown to be physically moving during drag and drop

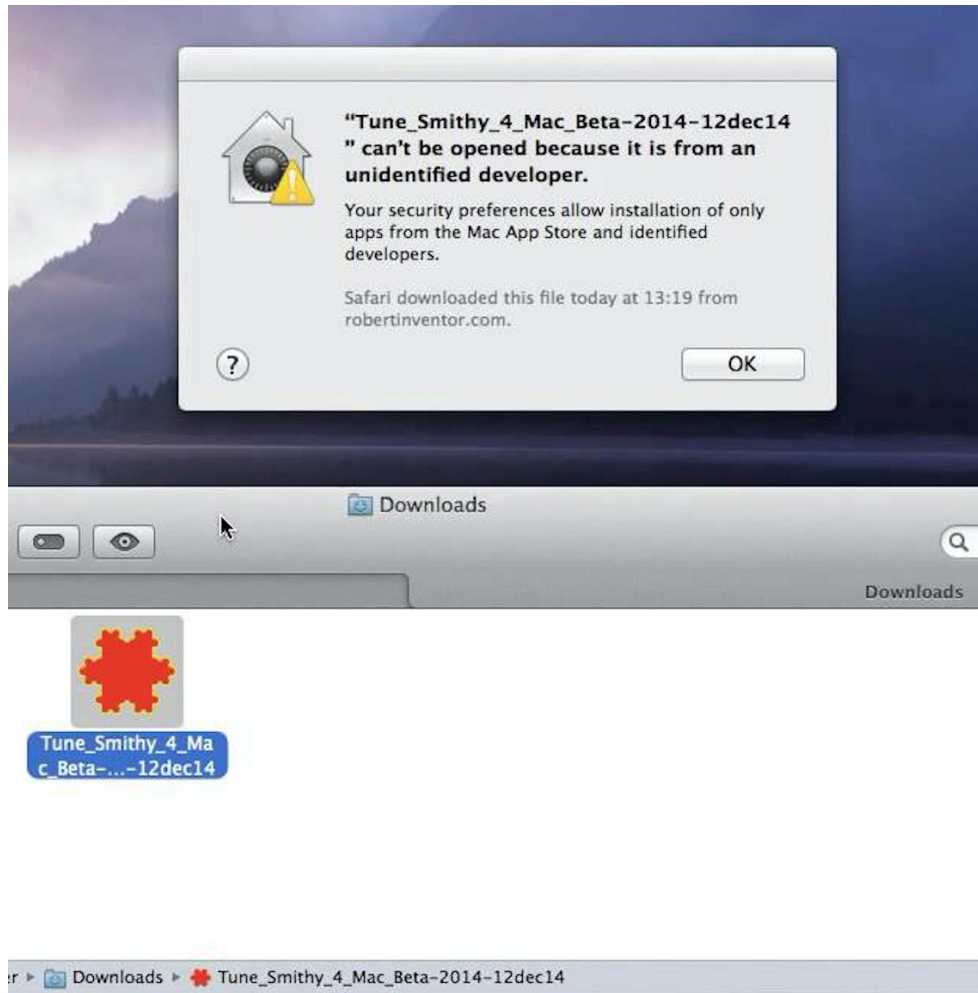
Have a clearly defined end-point in the interaction !

Principle (4/8) – Yield closure



Appropriate feedback must be given (For errors as well as others) !

Principle (5/8) – Error handling

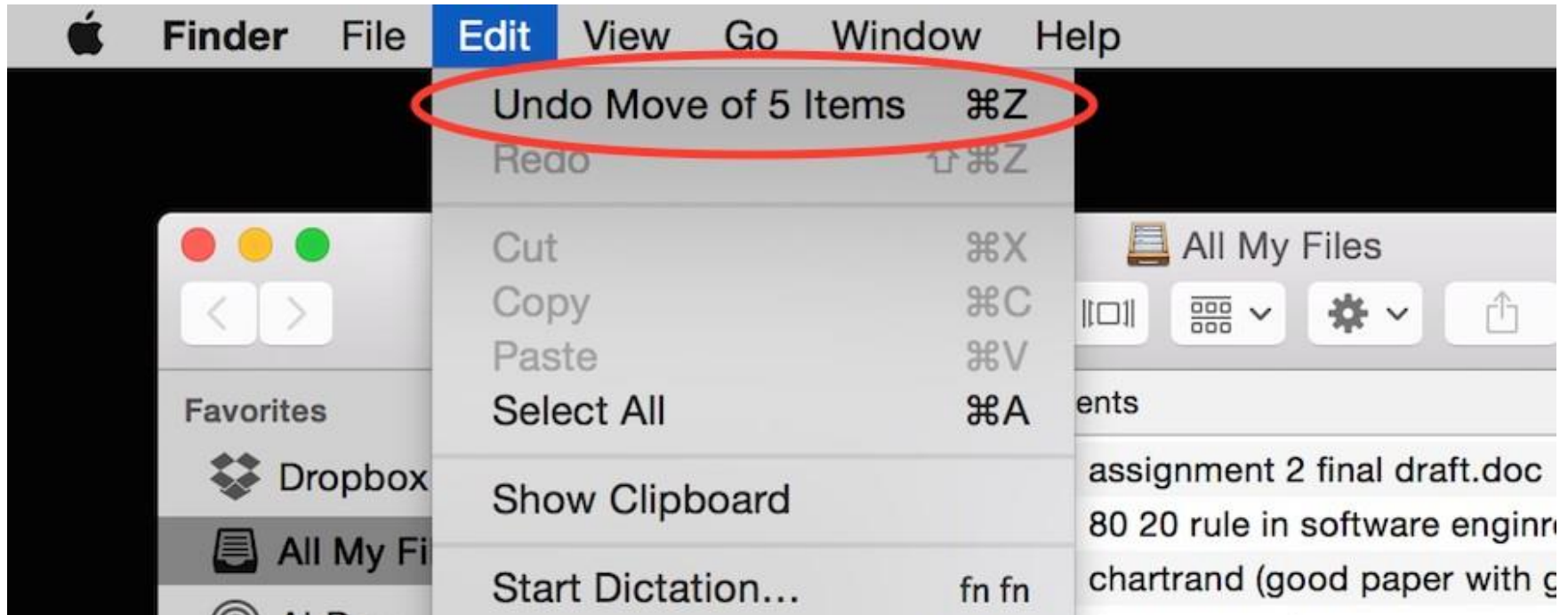


- Clearly tell users what was wrong
- Don't scare the users. Use appropriate language
- Don't just give Error handles or Error codes
 - “404 error” does not really convey much
- Only make users redo the part that was wrong

Principle (5/8) – Error handling

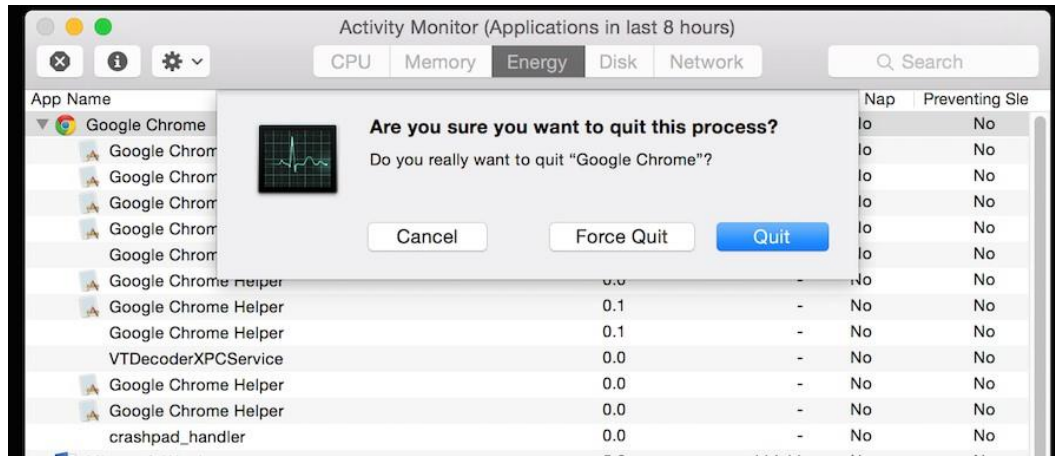


Principle (6/8) - UNDO

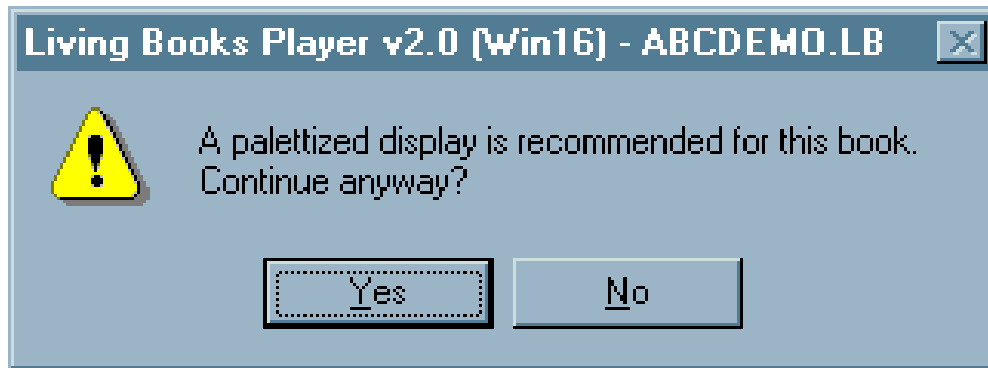


Actions must be reversible ! If some actions can't be undone, use “hesitation”

Principle (7/8) – Internal locus of control



- Inexperienced users may be intimidated when the software makes decisions
- Experienced users want to control the flow
- Put the users in-charge (to the extent possible)
 - Hide technical terms
 - Interaction must be customizable (Personalization)



Principle (8/8) – Short term Memory (STM)



iOS 4

- Remember 7 ± 2
- Apple has only 4 apps in the main menu
- Organizing things hierarchically help reduce STM

Usability Guidelines

Usability Guidelines are statements by which to determine an action of usefulness. A Usability guideline aims to streamline particular processes according to a set routine or sound practice. By definition, a guideline is never mandatory. Guidelines are not binding and are not enforced.

